

Article

Proposal of Restrictions on the Departure of Korea Small Fishing Vessel according to Wave Height

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Abstract: The Korean government imposes departure restrictions on fishing vessels of a gross tonnage of less than 15 tons under adverse weather conditions when significant wave heights exceed 3 m. However, an analysis of small fishing vessels indicates that frequent capsizing accidents occur even under the current departure restrictions of permission when significant wave heights are below 3 m. In this study, we propose new guidelines for regulating the departure of Korean small fishing vessels based on adverse weather conditions. To achieve this, we compare departure restrictions under weather deterioration for small fishing vessels in different countries worldwide with those in Korea. Additionally, by incorporating experimental research conducted in the United Kingdom on the capsizing of fishing vessels, we present a new departure restriction equation considering the ship characteristics (a narrow beam-to-length ratio) of Korean fishing vessels measuring 24 m or less. Through this analysis, we provide detailed guidelines for departure restrictions on Korean small fishing boats based on their length and wave height. To validate the proposed guidelines, we analyze fishing vessel capsizing accidents that occurred near the Korean coast over the past 23 years and demonstrate the utility of the new guidelines by applying them to past capsizing incidents.

Keywords: Wolfson unit; small fishing vessel; sailing restriction; Imo stability; wind wave advisory



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1. Introduction

Fishing workers operating on small vessels are vulnerable to maritime accidents. These vessels are primarily designed based on less stringent criteria, which is one of the main reasons for the vulnerability of these fishing vessels. To address this issue, international agreements such as the Torremolinos International Convention, Torremolinos Protocol, and Cape Town Agreement have been established to improve safety standards for fishing vessels over 24 m. However, for small fishing vessels of less than 24 m, there is a lack of established international safety standards, resulting in inadequate or non-existent national regulations, guidelines, or standards in many countries [1].

To improve the safety of small fishing vessels, early research in this field has focused on investigating the relationship between fishing vessel stability parameters and ship specifications to estimate the representative parameters of GM and the restoring moment. Previous studies have analyzed this relationship and examined the suitability of linear models for predicting stability performance according to IMO standards [2]. Conversely, there are studies that estimate GM using genetic programming based on ship specifications [3]. Based on the GM equation, the paper calculates the suitable width of a fishing vessel for safe operation by utilizing the depth-draft ratio [4].

Researchers have also focused on conducting safety assessments of ship operations, employing various tools to explain the factors associated with ship accident rates. Some of these tools include the calculation of the anticipated accident probability in each navigational zone based on weather conditions [5]. In other studies, the incident location, weather data, and ship specific information have been combined to estimate risk prediction scenarios and assess ship safety, considering variables such as wind speed and wave height [6].

Furthermore, regression trees have been employed in research to estimate the relative accident rate by predicting weather and ocean activity at specific times and locations, thereby evaluating the extent to which waves and weather factors contribute to a ship's listing [7].

Many researchers have utilized various parameters related to waves to discover various mechanisms that affect a ship's capsizing and have proposed safety warning indicators combining wave height and steepness. They predicted the potential risks of rogue waves [8]. Additionally, some reports [9] presented the critical wave height for capsizing based on ship specifications, wave height, and period. McCue [10] conducted a probabilistic analysis of the multi degree of freedom capsizing risk, while Francescutto [11] evaluated the stability performance of ships under rolling motions caused by long period waves. Furthermore, Guachamin-Acero [12] assessed safety when considering uncertainty factors such as significant wave height, wave period, weather forecasts, and wave spectrum. Moreover, Deshmukh [13] developed a small ship capsizing risk index considering wave steepness.

In Korea, the government has recognized the dangers of waves related to ships and has taken steps to address this issue by modifying its related regulations. Specifically, the Fishing Vessel Safety Management Act [14] of the Republic of Korea has been amended to impose tightened regulations on fishing vessels below a certain tonnage when wind wave advisories are issued. This is because there is a strong connection between navigating in rough seas and major maritime accidents that can result in significant human casualties, such as capsizing accidents.

Looking at the actual occurrence of maritime capsizing accidents with respect to weather warnings in Table 1, the number of capsizing accidents has been declining since 2020. However, it is important to note that since 2018, the number of capsizing accidents has continued to rise on days with weather conditions that do not meet the criteria for issuing weather warnings. A staggering 82% of all capsizing accidents that occurred between 2017 and 2022 happened on days when no weather warning was in place.

Table 1. Comparison of Capsize Incidents between Wind Wave Advisory and Non-Wave Advisory Days in Korea.

	Capsize Accidents on Days without Marine Weather Advisory ¹	Capsize Accidents on Days with Marine Weather Advisory	Total
2018	38	8	46
2019	78	32	110
2020	81	27	108
2021	89	7	96
2022	89	9	98
Total	375 (82%)	83 (18%)	458

* Source: Korea Maritime Safety Tribunal. ¹ Marine Weather advisory: The weather advisory, closely related to marine safety, is issued to warn of expected weather conditions that are likely to occur over a certain period.

Most of the weather warnings issued at sea are related to strong wind waves and typhoons. Assuming that the days without such warnings are under the condition of a wind wave advisory not in effect, it can be inferred that many accidents occur in waters with significant wave height of less than 3 m in South Korea. This highlights the importance of monitoring and taking preventive measures in areas with these sea conditions, as weather warnings are not always sufficient in preventing accidents.

Figure 1 classifies vessel accidents that occurred over the past five years since 2018 based on the size of the vessel in tons. Of the total 480 accidents, 89% or 428 occurred on ships with a total tonnage of less than 10 tons. Considering the large number of registered vessels under 10 tons in Korea, it is understandable that accidents occur more frequently. However, this highlights the need for safety navigation standards for small fishing vessels.

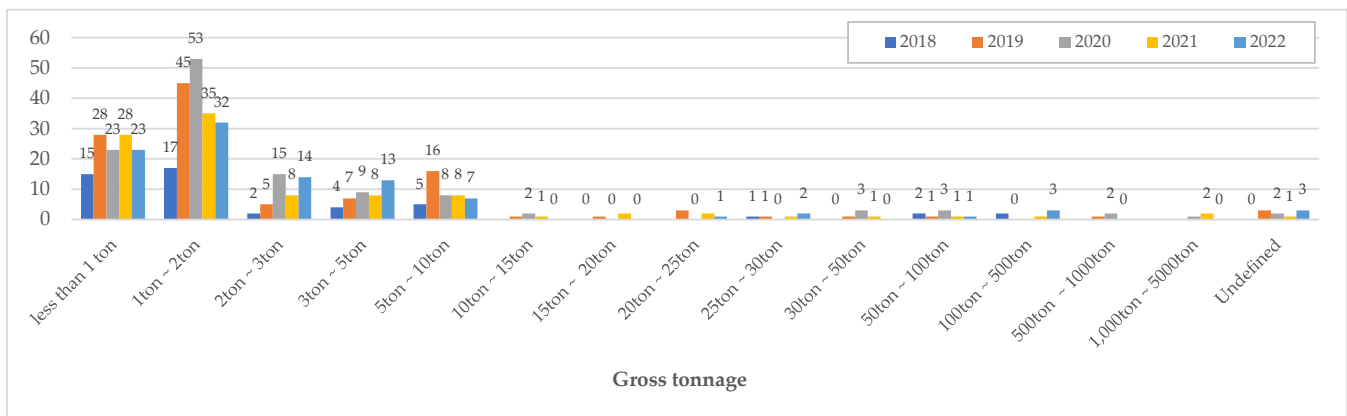


Figure 1. Classifying Capsizing Incidents of Korean Vessels by Tonnage (2018–2022). * Source: Korea Maritime Safety Tribunal.

To propose restrictions on fishing vessel departure according to wave height, accident data analysis should be carried out to determine the most common causes of accidents on fishing vessels [15]. The analysis revealed that typical causes include overloading on deck during bad weather, deck flooding during navigation, and capsizing due to shore waves [16]. Freeboard is an important factor in ensuring the stability of a ship, especially in rough seas or adverse weather conditions. In cases of capsizing, it was found that many accidents occurred when the deck was flooded, and the vessel maintained a list for a while and sank completely upon reaching the angle of inundation due to additional external forces [17]. The study investigated the probability of capsizing in irregular beam seas and found that it is more closely related to the decreased stability caused by deck flooding. Additionally, the study showed that the effect of freeboard height on the amount of trapped water on deck is more significant than that of the metacentric height, and the danger of capsizing is also affected by the rise of floor [18]. Furthermore, it is important to highlight that small boats are highly vulnerable to low wave heights, particularly when specific factors come into play, such as wave period, wave direction from the beam, ship width, and when the breaking wave height matches the beam height of the boat [19]. Considering these circumstances, the implementation of restrictions on departure based on wave height becomes critically significant, particularly for small fishing vessels.

Furthermore, small fishing vessels in Korea have a lower center of gravity due to their lightweight and small size, making them more susceptible to capsizing due to external forces such as waves and wind. In addition, Korean small fishing vessels are characterized by their narrow width. The ratio of beam-to-length is smaller compared to the EU's. As a result, they exhibit weaker reserve buoyancy and greater susceptibility to sudden waves and winds [2].

In addition to the stability performance determined by ship specifications, some fishermen have made unauthorized modifications to their vessels for various reasons, such as economic benefits. These modifications often impact vessel stability and put the vessel and its crew at risk. To enhance the safety of small coastal fishing vessels (less than 10 tons), researchers in Korea have established maximum length limits based on the vessel's tonnage [20]. It eventually increased vessel stability by improving the ratio of vessel length to width and depth, thereby enhancing the vessel's buoyancy and its stability.

Another measure to prevent capsizing accidents is the implementation of departure restrictions [14]. Currently, fishing vessels below 15 gross tons are prohibited from departing when the significant wave height exceeds 3 m. However, despite complying with these restrictions and operating under weather conditions with wave heights below 3 m, capsizing accidents still frequently occur among small vessels [21,22]. The insufficient effectiveness in preventing such accidents may be attributed to the inadequacy of the existing departure restrictions for Korean small fishing vessels. To compare this, we seek to explore

the legislation in other countries that recommend operational restrictions on fishing vessels and other small boats.

Moreover, this study aims to address this issue by quantitatively refining the current departure criteria and proposing new guidelines. Although there are numerous studies in the field of ship stability that analyze the causes of capsizing accidents, there is a lack of research dedicated to investigating departure criteria settings, particularly in terms of the specific and quantitative vulnerability associated with wave heights.

To achieve this objective, the departure restrictions for small Korean fishing vessels were derived based on capsizing experiments conducted at the Wolfson Unit in the UK and the intact stability criteria outlined in the 2008 IS Code [23]. These derived restrictions were developed to be suitable for vessels typically measuring under 24 m, taking into consideration the differences in ship specifications and characteristics between the vessels used in the experiments and those found in Korea. Furthermore, the formula utilized in deriving these restrictions possesses a non-dimensional nature, enabling its application to vessels of varying sizes. This aspect of the formula enhances its versatility, making it applicable to fishing vessels under 24 m in Korea.

By implementing the newly proposed departure guidelines, a higher level of safety can be achieved, especially in preventing capsizing accidents. These guidelines provide operators with essential safety measures, such as guidance on navigating hazardous wave conditions and issuing warnings about minimum wave heights. Adhering to these guidelines significantly improves the overall safety of small fishing vessel operations and effectively reduces the risks associated with adverse wave conditions. As a result, operators can enhance the safety of vessel operations by following reliable safety guidelines and taking appropriate preventive measures to minimize the occurrence of capsizing accidents.

2. Comparison of Regulation Related to Restriction on the Departure of Small Vessel including Fishing Vessel

This chapter discusses the departure restrictions and safety measures for vessels, particularly fishing vessels and small boats, operating in rough seas. The main objective of this study is to compare and analyze the current departure restriction regulations in Korea with the maritime navigation warning policies or cases implemented in other countries. Through this comparative analysis, the effectiveness of current regulations in Korea can be assessed, and areas for potential improvements can be identified.

2.1. Korean Fishing Vessel Departure Restriction Act

According to “The Enforcement Rules of the Act on the Safety Management of Small Fishing Vessel”, all vessels are prohibited from departing in weather conditions of a wind wave advisory or higher as shown in Table 2. When a wind wave advisory is issued which is one level lower than a wind wave warning is expected, vessels with a total tonnage of less than 15 tons are restricted from departure. If the ship has already departed and is in operation, it must immediately return to port.

Table 2. Criteria for restrictions on departure of Korean fishing vessels.

Weather Warning	Criteria for Effectuation	Vessels with Departure Restriction
Wind wave advisory	Sustained wind speed of 14 m/s or more or wave height of 3 m or more	Fishing vessels under 15 G/T Fishing vessels under 30 G/T in winter season
Wind wave warning	Sustained wind speed of 21 m/s or more or wave height of 5 m or more	All fishing vessels

Source: The Enforcement Rules of the Act on the Safety Management of Small Fishing Vessel.

To minimize accidents and human casualties during winter, from November to March, vessels with a gross tonnage of up to 30 tons are prohibited from departing when a wind wave warning is in effect. This restriction has been expanded to include fishing vessels

with 30 G/T or less, as they are predominantly under 24 m in length and were manufactured without compliance with the regulation for the approval of structural recovery performance during their design [24]. This makes it difficult for them to ensure safe navigational performance. As a result, relatively lenient departure restrictions have been imposed on Korea compared to other countries.

Meanwhile fishing leisure vessels are a type of fishing vessel that are subject to more stringent regulations compared to fishing vessels. A fishing leisure vessel is a small ship with a gross tonnage of less than 10 tons and a maximum passenger capacity of 22 people. Its primary purpose is to transport passengers to fishing grounds and facilitate fishing activities at sea. Fishing leisure vessels were subject to the same departure restrictions as other fishing vessels until 2018. However, due to frequent accidents causing numerous casualties, the Korean government revised the relevant regulations in 2019 to strictly enforce safety standards for fishing vessels, even restricting departures during weather preliminary warnings. Departures are restricted not only to wind wave warnings and advisories, but also when weather forecasts indicate sustained winds of 12 m/s or higher, or a significant wave height of 2 m or higher, as shown in Table 3.

Table 3. Criteria for restrictions on departure of Korean fishing leisure vessel Enforcement.

Weather Warning	Criteria for Effectuation	Vessels with Departure Restriction
Preliminary wind wave warning	Sustained wind speed of 12 m/s or more or wave height of 2 m or more	All fishing leisure vessels

Source: Decree of the Fishing Management and Promotion.

In summary, South Korea enforces restrictions on the departure of fishing vessels based on their gross tonnage. Fishing vessels with 15 gross tons or less are not permitted to leave the port when the maximum wave height exceeds 3 m. Moreover, all vessels are prohibited from departing during weather warnings. Specifically for fishing vessels which fall under the category of a fishing leisure vessel, the maximum allowable wave height is set at 2 m. This implies that for smaller fishing boats, if the wave height remains below 3 m, they are allowed to navigate regardless of the vessel's size, and there are no specific regulations in place.

2.2. Chinese Fishing Vessel Departure Restriction Act

The regulations regarding fishing vessel management in China vary slightly by region, but in general, the departure of fishing vessels is restricted based on the ship's engine output. Table 4 illustrates the departure restrictions applied to fishing vessels in the Shandong region of China. These regulations are divided into four main categories, assuming that ships with higher engine output are capable of navigating in rougher seas. According to these regulations, the navigation rating for fishing vessels, which includes the domestic or traditional way of catching fish, is gradually applied one level stronger based on the criteria Table 4, resulting in operational limitations.

Table 4. Criteria for restrictions on departure of Chinese fishing boats.

Wind Scale	Criteria for Effectuation	Vessels with Departure Restriction
Class 5	Average wind speed above of 8 m/s or wave height of 2 m	Non-motor fishing vessels
Class 6	Average wind speed above of 10.8 m/s or Wave height of 3 m	Fishing vessels with a power of less than 44 kw (60 ps)
Class 7	An average wind speed above of 13.9 m/s or Wave height of 4 m	Fishing vessels with a power of less than 298 kw (400 ps)
Class 8	An average wind speed above of 17.2 m/s or Wave height of 5.5 m	All fishing vessels

Source: Shandong Province Fishing Vessel Management Measures (Provincial Government Order No. 284).

In summary, the departure of Chinese fishing vessels is determined by detailed regulations based on the size of the ship's engine output. Based on wave height, only powerless vessels are allowed to depart in waves up to 2 m. In waves up to 3 m, the departure is restricted to small vessels equipped with 60 hp engines. In waves up to 4 m, the departure of vessels equipped with 400 hp engines are limited. However, in waves higher than 5.5 m, where the sea is extremely rough, the departure of all vessels is prohibited.

2.3. Safety Precautions for Small Fishing Boats in Case of Bad Weather in the U.S.

In the United States, weather forecasts for small craft vessels include a “small craft advisory”, and in some specific areas, a slightly lower level called the “exercise caution” is issued. These advisories are announced by the National Weather Service (NWS) and apply to all vessels that may be negatively affected by the specified weather conditions. The advisories do not provide a specific vessel size definition. When certain weather criteria are expected within the next 12 h, additional information is provided to vessel operators or captains to help them decide whether or not to set sail.

The criteria for issuing small craft advisories may vary by region, considering geographical characteristics and local weather patterns. Table 5 illustrates the conditions for issuing advisories in Florida. A “small craft advisory” is issued when forecasted waves are above 7 feet. A “small craft exercise caution” advisory, one level below an advisory, is issued when waves are at or around 6 feet or corresponding sustained winds are expected.

Table 5. Warning and Advisory Criteria of Melbourne, FL.

Weather Warning	Criteria for Effectuation
Small craft exercise caution	Sustained winds of 15 to 20 knots (7.7~10.3 m/s) or seas of 6 feet (1.8 m)
Small craft advisory	Sustained winds of 21 to 33 knots (10.8~17 m/s) or seas of 7 feet or greater (2.1 m)
Gale warning	Sustained winds of 34 to 47 knots (17.5 to 24.1 m/s)

Source: National Weather Service; <https://www.weather.gov/marine/faq> (accessed on 1 May 2023).

There are no official prohibitions or signs that explicitly forbid small vessels from navigating. However, in certain cases, local authorities or the Coast Guard may recommend refraining from sailing in a specific area until weather conditions improve. Additionally, some harbors or marinas may impose restrictions on small crafts, preventing them from leaving during high wind conditions.

To summarize the criteria for the aforementioned forecasts based on wave heights, a “small craft advisory” is typically issued when wave heights are expected to exceed 2.1 m, while a “small craft exercise caution” advisory is issued for wave heights below (1.8 m).

2.4. Safety Precautions for Small Fishing Boats in Case of Bad Weather in the Canada

In Canada, there are no specific regulations governing the operation of vessels, including small fishing vessels. However, the Canadian Coast Guard provides safety advisories to small craft operators as a source of weather information. These advisories are issued when wind speeds or wave heights have the potential to be hazardous for small crafts, and mariners are advised against sailing in such conditions. The responsibility for issuing these advisories or warnings lies with the Canadian Coast Guard as outlined in Table 6.

Table 6. Canada Warning/Advisory Criteria.

Weather Warning	Criteria for Effectuation	Safe Precaution
Small Craft Warning	Winds 20–33 knots excluding gusts. Wave heights 2–3 m	
Gale Warning	Winds 34–47 knots excluding gusts. Wave Heights 6–9 m	Those operating smaller vessels should avoid navigating in all marine areas

Source: Canadian marine warning program, Environment Canada's Meteorological Service of Canada.

A small craft warning is specifically issued when wind speeds range from 20 to 33 knots (excluding gusts) and wave heights range from 2 to 3 m. These warnings are primarily issued during the recreational or summer boating season and are intended for coastal or inland waters. They serve as important guidance for small craft operators, urging them to exercise caution and avoid navigating in these conditions to ensure their safety.

Furthermore, a “Gale Warning” is issued when wind speeds range from 34 to 47 knots (excluding gusts) and wave heights reach a range of 6 to 9 m. Operators of small vessels are strongly advised to avoid navigating in all marine areas, including coastal, inland, and offshore regions.

2.5. Comparison of Departure Restriction Criteria by Vessel Size between Countries

In this section, the comparison between the regulations regarding fishing vessel departure restrictions based on the previously analyzed deteriorating weather conditions and the regulations in Korea has been discussed. The summary of this comparison is presented in Table 7.

Table 7. Safety precautions for small fishing vessel and small craft as of wave height.

Country	Vessel	Criteria for Effectuation	
		Wind	Wave Height
Korea	Fishing leisure vessel	12 m/s or more (Sustained for 3 h)	2 m or more
	Fishing vessels less than 15 tons ¹	14 m/s or more (Sustained for 3 h)	3 m or more
	All fishing vessels	21 m/s or more (Sustained for 3 h)	5 m or more
China	non-powered fishing vessels	8 m/s or higher on average	2 m or more
	Fishing vessels with engine power less than 60 ps	10.8 m/s or higher on average	3 m or more
	Fishing vessels with engine power less than 400 ps	13.9 m/s or higher on average	4 m or more
	All fishing vessel	17.2 m/s or higher on average	5.5 m or more
U.S.	Small craft	Exercise caution *	seas of 6 feet (1.8 m)
		Advisory *	7 feet (2.1m) or greater
Canada	Small Craft	Warning Winds 20–33 knots on average, excluding gusts	Wave heights of 2–3 m

¹ The scope of the departure restriction is expanded to vessels under 30 tons in winter. * Thresholds governing the issuance of small craft advisories are specific to geographic areas.

According to Korean regulations, fishing leisure vessels are restricted from sailing when the wave height exceeds 2 m, while most common small fishing vessel face departure limitations for weather conditions with wave heights 3 m or more. The Chinese government stipulates that non-powered vessels require a minimum wave height of 2 m for departure, and it further categorizes the restrictions based on engine power, ranging from 3 m to 5.5 m. In the United States and Canada, departure restrictions are determined based on vessel size, with limits set between wave heights of 1.8 to 2.1 m and 2 to 3 m.

After comparing regulations in Korea with those of other countries targeting small vessels or fishing boats, it was observed that, unlike Korea, other countries not only con-

sider vessel size but also incorporate their own criteria, such as engine output or any condition that may adversely affect the stability of the vessel. As a result, small craft and small fishing vessels are subject to navigational warnings or restrictions to avoid navigation when the wave height reaches approximately 2 m. Given these findings, it appears necessary for the Korean government to consider reviewing the existing departure limitations for small fishing vessels with wave heights below 3 m during adverse weather conditions.

3. Cases of a Capsize Accidents of Small Fishing Vessels in Korea

This chapter aims to investigate the occurrences of capsize accidents. The data utilized in this study is derived from capsize accident reports provided by the Korea Maritime Safety Tribunal, spanning the period from 1999 to 2022. The primary focus of the analysis is to explore the correlation between the length of vessels and the wave height in capsize accidents primarily caused by sea waves. By utilizing accident information, including causes, vessel length, tonnage, and wave height at the time of each incident, our objective is to determine the frequency of accidents that occurred at lower wave heights in comparison to the conditions specified for departure restrictions. The findings of this research can potentially serve as a foundation for the revision of existing regulations governing departure restrictions, thus contributing to the enhancement of maritime safety in Korea.

Case of a Capsize Accidents of a Fishing Boat in Korea

In this section, we analyzed the current status of capsizing incidents in Korean fishing vessels using the Korean Maritime Accident Investigation Report [25]. We selected data from capsizing incidents classified under the causes of “waves” and “wave heights” that occurred in the Korean coast area from 1999 to 2022, a total of 23 years. The accident reports commonly indicated wave conditions as the main cause. To specifically focus on accidents caused by deteriorating weather conditions, incidents such as “capsizing accidents that occurred during operations involving crane or net lifting” were excluded from the analysis. As a result, a total of 66 cases of “fishing vessel capsizing incidents due to deteriorating weather conditions” were analyzed, and the results are presented in Table 8.

Table 8. Maritime Safety Tribunal Verdict Cases of Capsize Accident (1999–2022).

	LOA (m)	G/T	Wave Height		LOA (m)	G/T	Wave Height		LOA (m)	G/T	Wave Height
1	14.5	9.77	2.8	23	25.86	79	3.5	45	14.6	7.93	2.3
2	20.4	27	2.8	24	14.6	9.77	4.8	46	21.59	29	3.5
3	21.49	29	4.5	25	33.96	132	3.5	47	12.5	7.93	1
4	26.88	59	3	26	20.54	24	3.5	48	5.71	0.93	3
5	10.74	4.98	3.5	27	25.05	77	2.5	49	13.8	7.93	1.8
6	17.9	22	3	28	24.86	51	3.5	50	17.3	26	2.5
7	27	89.06	4	29	21.85	33	2.5	51	11.18	4.87	3.5
8	20.5	29	3.5	30	13.27	7.93	3	52	13.66	7.93	1
9	26	92.53	3.5	31	36.4	139	3.5	53	17.8	19	3.5
10	16.3	9.77	2.5	32	17.84	20	2.5	54	18.32	24	2
11	15.18	9.77	3	33	18.05	25	3	55	22.25	47	3.5
12	9.18	3.3	2	34	7.2	1.7	4.5	56	13.9	7.93	2.5
13	11.58	4.91	2.3	35	6	0.98	2	57	13.86	9.77	3
14	35.5	138	3.5	36	21.1	40	3	58	14.97	11	4.6
15	10.65	4.99	1.5	37	13.89	7.93	3	59	26.9	90.19	5
16	8.9	3.86	3.5	38	20.65	29	5.5	60	7.56	1.63	2.5
17	13.8	7.93	1.5	39	14.2	9.77	1	61	21.59	29	4
18	13.16	7.93	1.5	40	5.58	0.97	1	62	13.3	7.93	1.8
19	26	92.53	3.5	41	6.52	0.84	1.5	63	22.5	51.14	3.5
20	27.4	58	3	42	8.25	3.15	3	64	26.55	59	4
21	22.98	72	3	43	10.71	3.89	3.5	65	31.75	98	4.5
22	8.2	1.76	2	44	7.18	1.2	3	66	19.5	29	4.5

As seen in Table 8, the wave heights at the time of the accidents were analyzed based on the vessel length and gross tonnage. By utilizing this chart, it is possible to determine

the wave heights during capsizing incidents for small fishing vessels based on the ship size in the Korean coast area.

The analysis of wave height in fishing vessel capsizing incidents caused by deteriorating weather conditions (wave height) in the Korean coast area over the past 23 years is presented in Figure 2. The findings reveal that a significant number of capsizing incidents occurred even in wave conditions lower than the “fishing vessel departure restriction criteria” with a wave height of 3 m or more. Furthermore, incidents of capsizing were observed even in environments with wave heights of 1 m. The fishing vessel specifications for the capsizing incidents ranged from 5.6 m to 36.4 m in vessel length, with an average length of 17.6 m. Currently, the vessel size criterion for the application of departure restriction regulations is set at 15 G/T. However, according to the analysis of accident data, 33 incidents, accounting for 50% of all capsizing incidents, occurred in vessels less than 15 G/T.

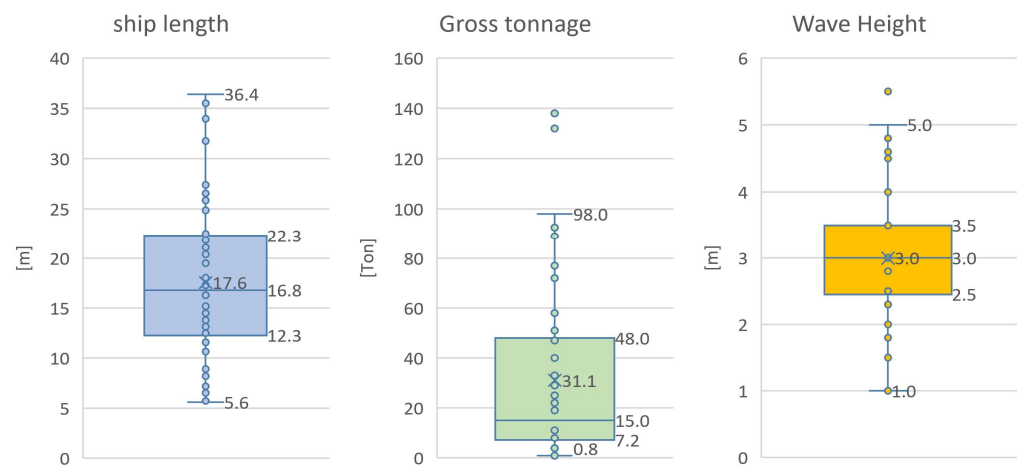


Figure 2. L.O.A., Gross Tonnage and Wave height of Capsizing Accidents in the period from 1999–2022.

The capsizing incidents presented in Table 8 are classified according to vessel specifications and visualized in Figure 3. Figure 3a illustrates the relationship between vessel length and wave height at the time of the accidents. The “Wind wave advisory criteria” (red dashed line) represents the departure restriction regulations for small fishing vessels. This criterion applies to fishing vessels with 15 Gross tons and restricts the vessel departure in wave conditions of 3 m or more. Therefore, in weather conditions with significant wave heights below 3 m, the decision to depart is subject to the operator’s subjective judgment. As seen in the figure, out of the total 66 capsizing incidents, 25 incidents occurred under wave heights of less than 3 m, accounting for 38% of all incidents. Including incidents that occurred under significant wave heights of 3 m, a total of 38 incidents out of 66 occurred under wave heights of 3 m or less, representing 58% of all capsizing incidents.

Figure 3b represents the total 25 incidents that occurred under wave heights of less than 3 m, focusing on vessels with less than 15 G/T as depicted in Figure 3a. As seen here, capsizing incidents occurred even under wave conditions of less than 3 m for small fishing vessels weighing less than 15 G/T. Additionally, there are six incident cases shown in relatively larger vessels with a gross tonnage of 15 tons or more.

Considering these incident cases, it is necessary to expand the current departure restriction criteria of “significant wave heights of 3 m or more” to include more detailed regulations for significant wave heights of 3 m or less. Furthermore, it is deemed necessary to establish new operational restriction criteria based on beam or length, which are closely related to vessel stability, to prevent capsizing incidents in small fishing vessels due to deteriorating weather conditions.

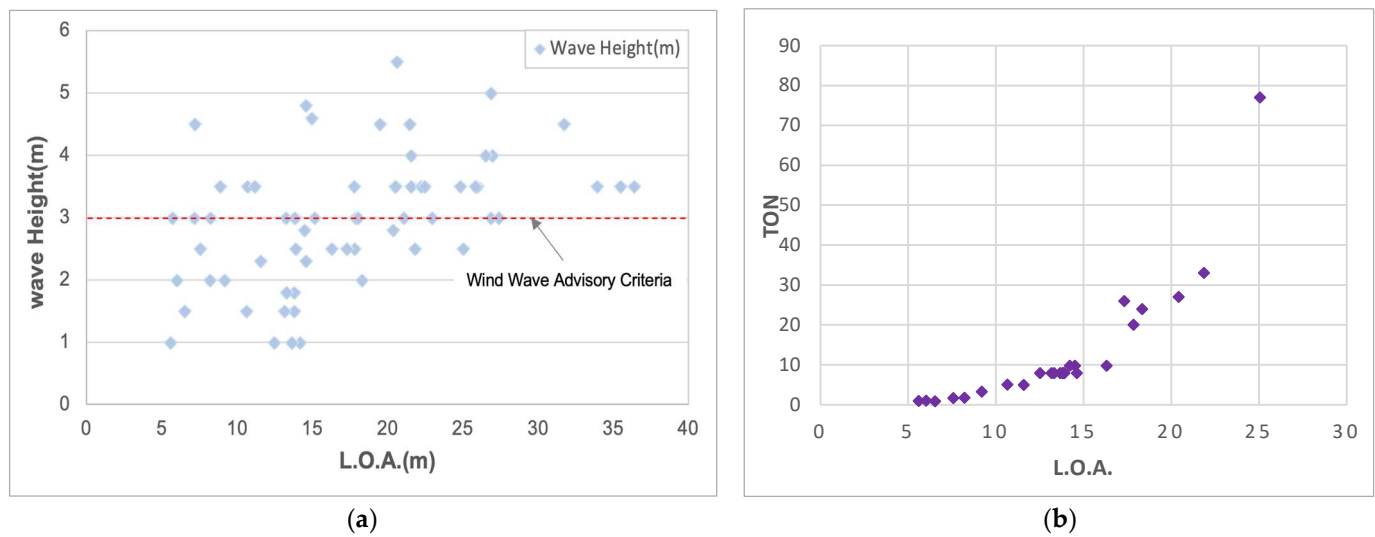


Figure 3. (a) Correlation between wave height and vessel capsizing occurred from 2000 to 2022; (b) the relationship between the tonnage and length of fishing vessels that capsized in waves with a height of less than 3 m in Korea.

4. Guidelines for Restrictions on the Operation of Small Domestic Fishing Vessels

The objective of this chapter is to propose a derived formula for suggesting departure restrictions for fishing vessels based on wave height. The aim is to propose a wave height formula for Korean fishing vessels, considering their appropriate vessel lengths. This proposal will provide suitable operational criteria for departure restrictions of fishing vessels.

4.1. MCA Research Project

In this section, we aim to utilize the research findings from the MCA research projects [19,26,27] as the basis for proposing departure restriction regulations for small fishing vessels in Korea. The aforementioned research has derived the minimum wave height at which a vessel capsizes in UK by assessing the stability performance of various model ships, particularly in response to capsizing.

According to this study, ship capsizing accidents are closely associated with the properties of the righting lever curve, particularly the range of the positive stability angle, maximum righting moment, and beam. These three components are utilized in Equation (1) to predict the “minimum capsizing wave height (H_{crit})” where *Range*, *RMmax*, and *B* represent the residual range of the positive stability in degrees, the maximum residual moment, and the maximum beam in meters, respectively.

$$H_{crit} = \frac{Range \sqrt{RMmax}}{10B} \quad (1)$$

Furthermore, the concept of the “operational limiting sea state” (Hs_{crit}) is introduced as followed in Equation (2). It is defined as half the height of H_{crit} according that it takes into consideration the potential encounter with twice the significant height waves. This concept allows for an intuitive assessment and utilization of the sea state [26].

$$Hs_{crit} = \frac{Range \sqrt{RMmax}}{20B} \quad (2)$$

Due to the necessity of estimating the dangerous wave height, even in situations where operators are unaware of the specific intact stability parameters outlined in the 2008 IS code, the researchers utilize a GZ value of 0.2 m for the maximum righting moment and estimate a range of positive stability angle at 45 degrees in the Hs_{crit} equation. This approach allows them to determine the safe wave heights for simulated database vessels, as

illustrated in Figure 4. As observed, Equation (3) describes the sea state in which a vessel could be vulnerable if it merely adheres to the minimum requirements of the IMO intact stability criteria.

$$Hs_{imo} = \sqrt{1+0.4LOA}-1 \quad (3)$$

where LOA represents the overall length of the vessel.

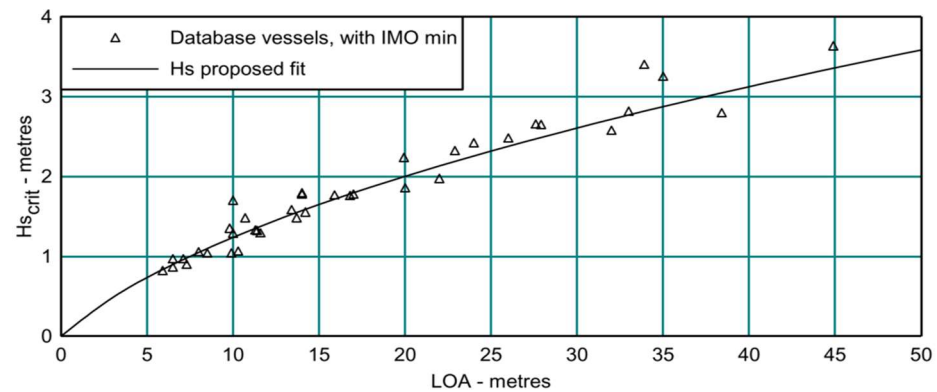


Figure 4. Variation of Hs_{crit} with length, assuming stability is equal to the IMO minimum criteria.

The Equation (3) represents the derivation of the minimum capsizing wave height for ships, considering the length and width trends observed in the UK experiments, which led to the proposal of the concept of significant wave height. In this study, we aim to utilize this concept to establish criteria for restricting the departure of small fishing vessels in Korea based on dangerous wave heights. However, due to the differences in specifications and characteristics between small fishing vessels operating in Korea, and the ships used in the MCA research project, it is necessary to consider these factors to propose new departure criteria that are suitable for Korea.

4.2. Safety Guideline for Korean Fishing Vessel

In this section, the focus is to propose a new formula suitable for Korean small fishing vessels based on the previously introduced Hs_{imo} formula, which was derived in the previous chapter. To achieve this, a comparative analysis is conducted between Korean small fishing vessels and the ship specifications employed in previous studies, considering the UK ship specifications as a reference for the new proposal.

The existing studies were derived based on UK ship specifications, but there are limitations to directly applying them to Korean fishing vessels. This is because the L/B ratio of Korean ships differs from the ship data used in the UK study. This difference is because the capsizing risk, which represents the stability of a ship, is correlated with the width of the ship, and Korean fishing vessels, which have narrow widths compared to their lengths, have higher capsizing risks. Therefore, we intend to modify the formula accordingly to account for the difference in width [2]. Thus, in this study, we compare Korean small fishing vessels and the ship specifications used in previous studies to propose a new formula suitable for Korean small fishing vessels.

The aforementioned UK ship estimation formula used 42 ships with a length of 24 m or less from the Wolfson wave height experiment. Table 9 presents the data used in the estimation formula for Korean small fishing vessels. This data is based on the small fishing vessel data (less than 24 m) publicly available as of June 2022 from the Korea Maritime and Port Safety Agency (KOMSA).

Table 9. Overview of the length of small vessels less than 24 m by gross tonnage.

	Number of Vessels	Min	Average	Max
Less than 1 ton	11,111	2.0	5.5	9.5
1~2 ton	20,702	3.9	6.9	11.3
2~3 ton	7840	2.8	8.3	11.7
3~5 ton	10,817	3.1	9.7	15.4
5~10 ton	9389	5	13.4	21
10~15 ton	708	11.8	16.3	20.8
15~20 ton	499	8.18	17.5	22.5
20~25 ton	543	14.55	20.1	23.9
25~30 ton	493	16.3	21.3	23.95
30~50 ton	400	18.35	22.6	23.99
50~80 ton	62	19.76	23.1	23.96
Total	62,564		13.7	

Source: Korea Maritime Transportation Safety Authority.

Table 10 summarizes the ship specification characteristics of Korean small fishing boats in Table 9 and those used in previous study [26].

Table 10. Comparison of the specifications of small vessels over 10 m and under 24 m in Korea with model capsizing tests in UK.

	KOREA		The UK	
	Range (m)	Average (m)	Range (m)	Average (m)
Length	2.0~23.99	13.7	6.5~23.9	13.6
Breadth	1.0~16.88	4.0	2.2~7.01	4.8
L/B		3.45		2.8

According to this, the average length was similar for both Korean and UK ships, at 13.7 m and 13.6 m, respectively. However, there was a difference of 0.8 m in the average width. Furthermore, the length-to-width ratio showed that Korean fishing vessels have a narrower and sleeker shape compared to UK ships, with a ratio of 3.45 for Korean vessels, as opposed to UK ships. This suggests that Korean small fishing vessels tend to have weaker stability compared to those of the UK, as Korean vessels in the same length category tend to be narrower and longer.

Figure 5 represents the length-to-width estimation formulas for Korean fishing vessels and the UK ships of the reference paper. The first graph depicts the estimation formula for Korean fishing vessels, showing that the width varies with the length according to the equation $B = 0.2975 L$. The second graph represents the length-to-width graph used in the reference study [19,26], where the corresponding estimation formula is $B = 0.3367 L$.

The comparison between the beam-to-length ratios of Korean domestic fishing vessels under 24 m in length and the ships used in the UK research is presented as follows:

$$B_{KOR} = 0.2975 * L \quad (4)$$

$$B_{U.K.} = 0.3367 * L \quad (5)$$

where L represents the length of the vessel and B_{KOR} represents the breadth of the vessel according to the ship's length in Korea and UK. The equations presented in Figure 5 demonstrate the relationship between the width and length of vessels. Notably, it is observed that Korean fishing vessels have a width that is approximately 12% less than their corresponding length, in comparison to the UK ships. This means that directly applying the existing UK formula to Korean fishing vessels would result in underestimating the significant wave height.

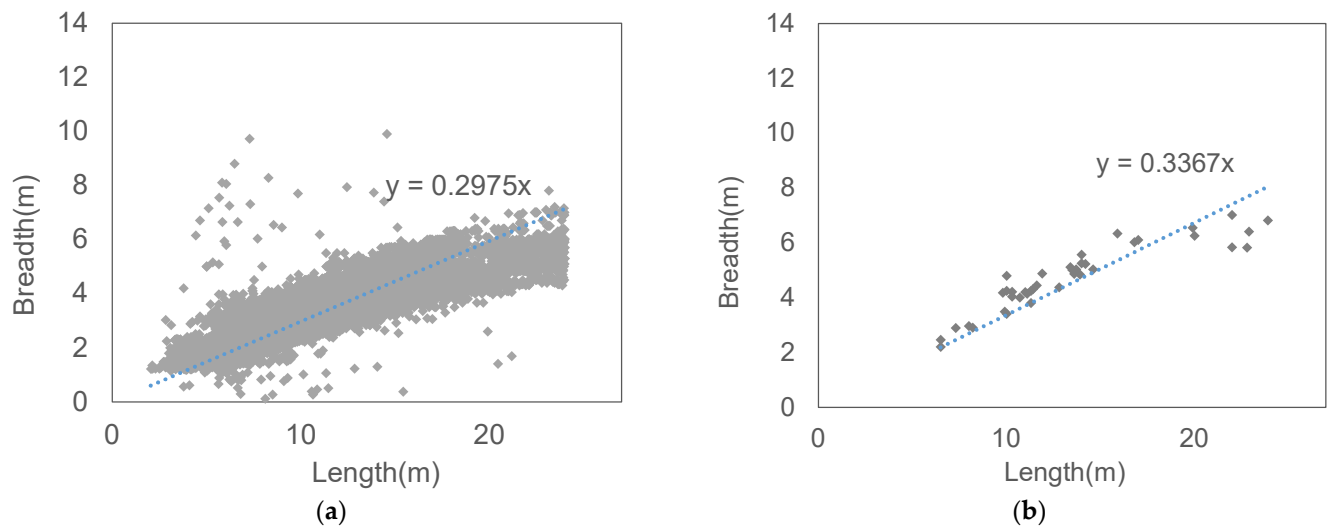


Figure 5. Width versus length of Korea fishing vessels and UK vessels. (a) Korea domestic fishing vessel under 24 m length. (b) The ship used in the former UK research.

Therefore, a new formula, denoted as Hs_K_{IMO} , has been proposed specifically for estimating the significant wave height of Korean small fishing vessels, considering the characteristics of both Korean and UK ships. Incorporating a coefficient of 0.88, which considers the length-to-width difference observed in Korean fishing vessels, revised formulas for estimating the significant wave height of small fishing vessels are presented as Equations (6) and (7).

$$Hs_K_{IMO} = \sqrt{1 + 0.4(0.88LOA)} - 1 \quad (6)$$

$$LOA = 2.841 * Hs_K_{IMO}^2 + 5.682 * Hs_K_{IMO} \quad (7)$$

where Hs_K_{IMO} means the minimum height of where a vessel may be at risk if it only meets the minimum requirements of the IMO intact stability criteria.

Equation (6) is derived to reflect the result that, compared to UK vessels of the same width, Korean vessels have a length that is 0.88 times shorter. This equation can be used to predict the significant wave height that leads to capsizing depending on the length of Korean small fishing vessels. Additionally, to determine the length of vessels subject to departure restrictions based on the sea state, a formula is needed that represents the relationship between the vessel's length and the sea state. Equation (7), which is a modification of Equation (6), represents the length of a ship exposed to risks based on wave height and can serve as a criterion for determining the sailing conditions of small fishing vessels depending on wave heights. By employing this formula, the departure restrictions for Korean small fishing boats can be categorized according to vessel length, taking into consideration weather conditions such as wave height.

4.3. Verification of a New Formula for Fishing Boat Capsize Accidents in Korea

In this chapter, the validation of the significant wave height formula (6) for Korean small vessels is performed. For this purpose, the effectiveness of the newly proposed formula for small vessel departure recommendations, using the dataset of 66 capsizing accidents in Korean vessels introduced in Section 3, is evaluated. The results are presented in Figure 6.

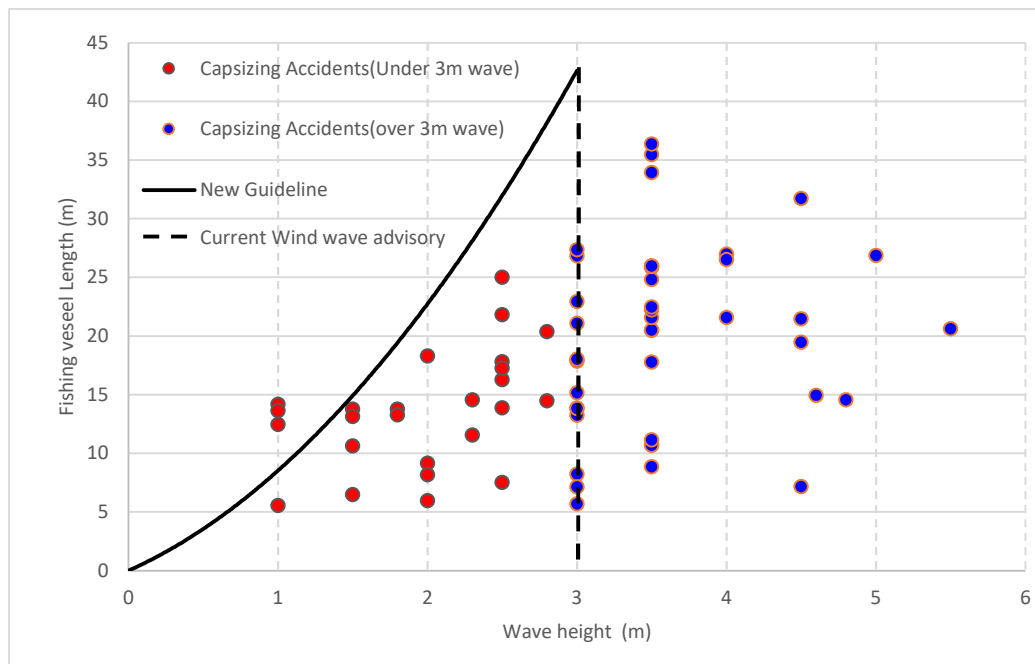


Figure 6. Comparison between the Korea Domestic Hs_KIMO Formula and Capsize accident data.

To evaluate the effectiveness of the newly proposed formula for small vessel departure recommendations, Figure 6 demonstrates that even when ships comply with the minimum level of existing departure guidelines, as represented by the dashed line indicating wind wave advisory in Korea, capsizing accidents (highlighted with red circles) continue to occur. In contrast, the evaluation of the newly proposed guidelines, depicted by the solid line, reveals that out of the 25 accidents indicated by the red circles occurring at significant wave heights below 3 m, a total of 22 accidents could have been prevented by adhering to the suggested new guidelines.

Furthermore, detailed guidelines, designed according to the ship's length, are presented in Table 11. These guidelines are based on Equation (7). Their purpose is to regulate the departure conditions of small fishing vessels in Korea in response to adverse weather. By utilizing these guidelines, it is anticipated that more specific departure conditions for small vessels can be established in accordance with varying weather conditions in Korea. Currently, the existing regulations impose departure restrictions for vessels under 24 m in length under weather conditions with wave heights exceeding 3 m. However, as shown in Figure 6, frequent capsizing accidents occur even in wave heights below 3 m for small fishing vessels in Korea. Therefore, it is expected that the implementation of more detailed departure restrictions will contribute to more effective standards for vessel safety and operation, particularly in mitigating capsizing accidents of small fishing vessels along the Korean coast.

Table 11. Guidelines of restrictions on the departure of small fishing vessels according to wave height.

L.O.A.	Wave Height	L.O.A.	Wave Height
2 m	0.31 m	14 m	1.43 m
4 m	0.55 m	16 m	1.58 m
6 m	0.76 m	18 m	1.71 m
8 m	0.95 m	20 m	1.84 m
10 m	1.13 m	22 m	1.96 m
12 m	1.29 m	24 m	2.07 m

Table 12 can also serve as a practical management tool for the authority. In accordance with the table, when the weather conditions reach a significant wave height of 1 m or above, fishing vessels with a length less than 10 m should be subject to departure restrictions or cautions. Likewise, in weather conditions with a significant wave height of 2 m or above, fishing vessels with a length less than 24 m should be restricted or cautioned from departing. Moreover, if the authority incorporates these guidelines as navigational information, issuing warnings or providing operators based on them, it will facilitate the practical application of the theoretical framework proposed in this paper.

Table 12. New management departure guideline of small fishing vessels according to wave height.

L.O.A.	Wave Height
10 m	1.0 m
24 m	2.0 m

5. Conclusions

In this study, new departure criteria were proposed for small vessels (under 24 m) operating in Korea based on weather conditions. To achieve this, the adverse weather departure conditions for small vessels in different countries worldwide were analyzed and compared with the existing departure regulations in Korea. Furthermore, departure guidelines specifically designed for small vessels in Korea were presented by utilizing experimental research on the significant wave height associated with the risk of the vessel capsizing. The validation of the proposed departure guidelines was conducted by analyzing capsizing accidents of fishing vessels that occurred along the Korean coast over a 23-year period starting from 1999. Applying the guidelines, it was observed that a total of 22 accidents out of 25 capsizing accidents with wave heights below 3 m, which occurred along the Korean coast, could be prevented. Additionally, the newly suggested departure guidelines according to vessel length will offer valuable information to enhance the safety of small fishing vessel operations along the Korean coast. This study has been validated based on past incidents of capsizing accidents and is expected to serve as a fundamental reference for recommending navigation restrictions in response to adverse weather conditions for small fishing vessels in Korea.

6. Limitation and Future Research

Three limitations can be identified in this study. Firstly, it should be noted that while we examined the applicability of the equation using wave heights at the time of accidents in Korea, there may exist other factors beyond waves that have not been considered. Secondly, we did not develop a direct formula using ship models representing Korean vessels, and the sample size of capsize experiments in the UK was limited. Lastly, it is important to acknowledge that for Korean vessels with a length of less than 24 m, adherence to the stability criteria specified by the IMO (International Maritime Organization) is not mandatory. As a result, if these vessels do not meet the criteria based on the G_z parameter, they could potentially capsize at wave heights lower than the derived critical capsize wave height which is specific to Korea.

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